

Travel Package Purchase Behaviour Analysis

1. Project Overview

This project analyzes customer purchase behaviour for travel packages offered by Trips & Travel Pvt. Ltd. The company offers five holiday packages — Basic, Standard, Deluxe, Super Deluxe, and King — to customers through a dedicated sales team. Despite active outreach, the overall conversion rate stands at only 18.8%, indicating that 8 in 10 leads do not result in a purchase.

The goal of this analysis is to identify the customer characteristics, sales pitch factors, and product preferences that drive purchase decisions, in order to help the sales and marketing team improve targeting, optimize pitch strategies, and increase conversion rates.

2. Dataset Summary

- Rows: 4,888
- Columns: 20
- Key Features:
 - Customer Demographics: (Age, Gender, MaritalStatus, Occupation, CityTier)
 - Financial Info: (MonthlyIncome)
 - Sales Interaction: (TypeofContact, DurationOfPitch, NumberOfFollowups, ProductPitched)
 - Travel Behaviour: (NumberOfTrips, Passport, PreferredPropertyStar, NumberOfChildrenVisiting)
 - Target Variable: (ProdTaken — 1 = Purchased, 0 = Not Purchased)
 - Missing Data: Age, DurationOfPitch, NumberOfFollowups, PreferredPropertyStar, NumberOfTrips, NumberOfChildrenVisiting, MonthlyIncome, TypeofContact

3. Exploratory Data Analysis using Python

I started with data preparation and cleaning in Python

- **Data Loading:** Imported the dataset using Pandas.
- **Initial Exploration:** Used `df.info()` to check the structure and `.describe()` for summary statistics.
- **Data Correction:** Standardized gender entry — replaced 'Fe Male' with 'Female'.
- **Missing Data Handling:** Used context-aware grouped imputation to fill missing values — Age filled using median grouped by Designation; TypeofContact filled with mode; DurationOfPitch, NumberOfFollowups and PreferredPropertyStar filled using median grouped by ProductPitched; NumberOfTrips filled using median by Occupation and Passport; NumberOfChildrenVisiting filled using median by MaritalStatus; MonthlyIncome filled using median by Designation.
- **Outlier Treatment:** Detected outliers using boxplots and IQR. Applied capping to DurationOfPitch and NumberOfTrips using IQR bounds. Applied percentile capping (1st–99th) to MonthlyIncome due to heavy skewness.

- **Data Type Correction:** Converted NumberOfFollowups, PreferredPropertyStar, NumberOfTrips, and NumberOfChildrenVisiting to integer type after imputation.
- **Column Standardization:** Renamed all columns to snake_case using regex for consistency and documentation readability.

4. Data Analysis — EDA (Univariate, Bivariate, Multivariate)

I performed a structured three-level EDA to understand feature distributions, relationships with the target variable, and combined interaction effects.

4.1 Univariate Analysis

Analyzed each feature independently using histograms with KDE for numerical features and count plots for categorical features.

- **Age:** Near-normal distribution. Majority of customers fall between 25–45 years.
- **MonthlyIncome:** Right-skewed. Most customers earn between INR 15,000–30,000 monthly. Outliers capped at the 99th percentile.
- **DurationOfPitch:** Most pitches last 8–20 minutes. Near-normal post-capping.
- **NumberOfFollowups:** Near-normal. Most customers received 3–5 follow-up calls.
- **ProdTaken (Target):** Highly imbalanced — 81.2% (0) vs 18.8% (1). Class imbalance is significant.
- **ProductPitched:** Basic package is most frequently pitched, followed by Deluxe and Standard.
- **MaritalStatus:** Married customers form the largest group, followed by Single and Unmarried.
- **Occupation:** Salaried employees dominate at approximately 60%+ of all customers.

4.2 Bivariate Analysis

Analyzed each feature against the target variable ProdTaken to identify purchase drivers.

Numerical Features vs ProdTaken — Median Comparison:

Feature	Not Purchased (Median)	Purchased (Median)	Difference
Age	36.0 years	33.0 years	Buyers are younger by 3 years
MonthlyIncome	INR 22,861	INR 21,137	Buyers have slightly lower income
DurationOfPitch	13.0 minutes	15.0 minutes	Buyers experienced longer pitches

Categorical Features vs ProdTaken — Conversion Rates:

Feature	Highest Converting Category	Conversion Rate	Lowest Converting
ProductPitched	Basic	29.97%	Other (6.99%)
MaritalStatus	Single	33.19%	Divorced (13.05%)
TypeofContact	Self Enquiry	21.85%	Company Invited (17.58%)
Occupation	Salaried	~20%	Other

4.3 Multivariate Analysis

- **Correlation Heatmap:** DurationOfPitch showed the strongest positive correlation with ProdTaken. Age and MonthlyIncome showed slight negative correlation.
- **Cross-Tabulation:** Customers who were Single + Basic product pitched + Self Enquiry showed the highest combined conversion — identifying this as the highest-value customer profile.
- **Stacked Bar Analysis:** Basic product consistently outperformed all other products across every marital status group.

5. Hypothesis Testing

Statistical tests were conducted to validate whether patterns observed in EDA are statistically significant and not due to random chance. Significance threshold was set at $\alpha = 0.05$.

5.1 Independent Sample T-Test (Numerical Features)

Two-sample T-tests were performed to check if the mean difference between buyers and non-buyers is statistically significant.

Feature	Test	P-Value	Result	Interpretation
Age	T-Test	$p < 0.05$	Significant	Buyers are statistically younger
MonthlyIncome	T-Test	$p < 0.05$	Significant	Income level differs significantly
DurationOfPitch	T-Test	$p < 0.05$	Significant	Pitch duration drives purchase

5.2 Chi-Square Test of Independence (Categorical Features)

Chi-Square tests were performed to check if categorical features have a statistically significant association with ProdTaken.

Feature	Test	P-Value	Result	Interpretation
TypeofContact	Chi-Square	$p < 0.05$	Significant	Contact type affects purchase

ProductPitched	Chi-Square	$p < 0.05$	Significant	Product type is a purchase driver
MaritalStatus	Chi-Square	$p < 0.05$	Significant	Marital status predicts purchase
Occupation	Chi-Square	$p < 0.05$	Significant	Occupation is associated with purchase

All tested features showed statistically significant relationships with ProdTaken, validating that the EDA patterns are reliable and not random.

6. Business Insights

- 1. Overall Conversion Rate is Low:** Only 18.8% of customers purchased a package — 8 in 10 contacted leads do not convert. The company needs smarter targeting rather than broad outreach.
- 2. Pitch Duration Matters:** Customers who purchased experienced a median pitch of 15 minutes vs 13 minutes for non-buyers. Longer, more engaging pitches are directly linked to higher conversion.
- 3. Basic Package Drives the Highest Conversion:** With a 29.97% conversion rate, the Basic package significantly outperforms Standard (16.71%) and Deluxe (11.78%). Customers prefer entry-level, affordable options.
- 4. Single Customers Convert 2.5x More:** Single customers convert at 33.19% — nearly 2.5x higher than Married (13.93%) or Divorced (13.05%) customers. They represent the highest-value segment.
- 5. Younger Customers Show Higher Purchase Intent:** Purchasing customers are younger (median 33 vs 36) and slightly lower income (INR 21,137 vs 22,861) — suggesting younger, budget-conscious customers are more open to travel packages.
- 6. Self Enquiry Leads Convert Better:** Self Enquiry customers convert at 21.85% vs 17.58% for Company Invited — inbound leads show stronger purchase intent.

7. Dashboard on Power BI

Built an interactive two-page Power BI dashboard to present key findings visually, enabling the sales and marketing team to explore customer behaviour patterns using filters and slicers.

Dashboard 1 — Conversion Overview & Customer Profile

Travel Package Purchase Analysis Dashboard - Executive Summary

4.89K

Total Customers

920

Total Purchases

37

Average age

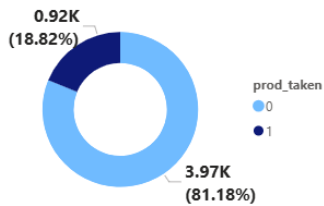
\$23.57K

Average Income

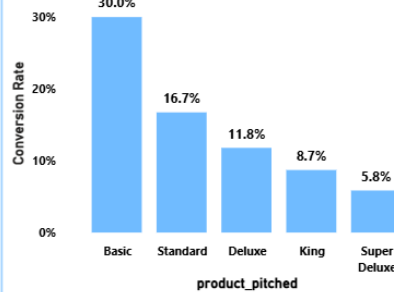
19%

Conversion Rate

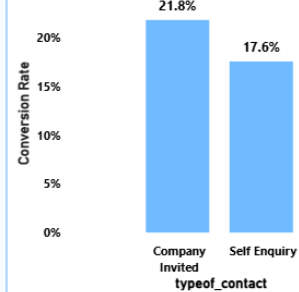
Purchase vs Non-Purchase



Conversion Rate by product_patched



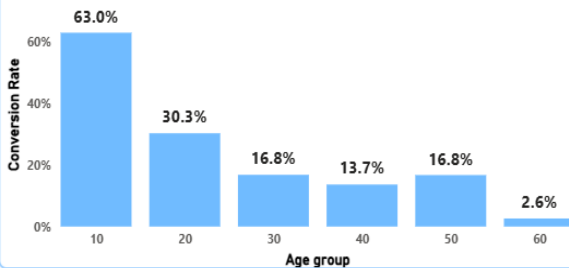
Conversion Rate by typeof_contact



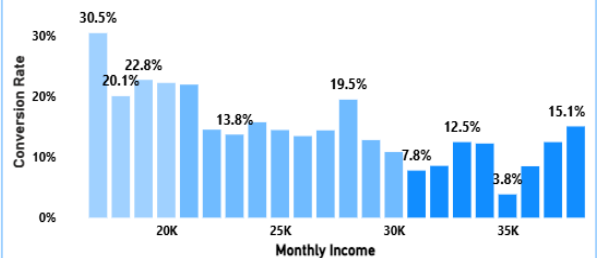
Dashboard 2 — Sales & Customer Segment Analysis

Travel Package Purchase Analysis Dashboard - Customer Analysis

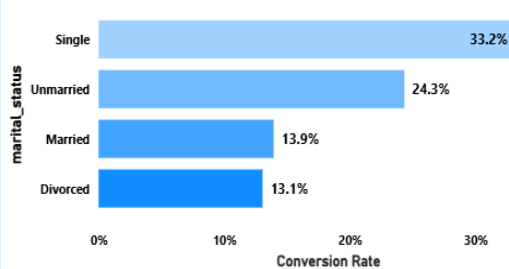
Age vs Conversion Rate



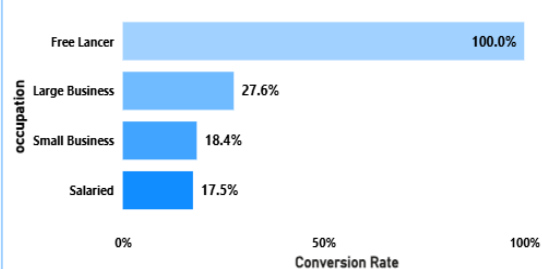
Conversion Rate by Monthly Income



Conversion Rate by marital_status



Conversion Rate by occupation



8. Business Recommendations

- Lead with Basic Package — Since Basic has the highest conversion rate (29.97%), prioritize it in first-time customer pitches and introduce upselling only after the initial purchase commitment.

- Train Sales Team on Pitch Duration — Buyers engage for 15 minutes on average. Develop structured pitch scripts that maintain engagement beyond 13 minutes to improve conversion.
- Target Single and Unmarried Customers — These segments convert at 33.19% and 24.34% respectively. Run dedicated campaigns with solo travel and group adventure packages for these groups.
- Design Age-Specific Promotions — The 28–35 age group shows highest purchase intent. Introduce EMI options, early-bird pricing, and budget-friendly packages for this segment.
- Invest in Inbound Marketing — Self Enquiry customers convert at a higher rate. Increase SEO, social media, and referral programme investment to generate more inbound leads.
- Optimize Follow-Up Strategy — Analyse optimal follow-up frequency per segment to avoid over-contacting low-intent leads while maintaining engagement with high-potential customers.
- Focus Outreach on Salaried Professionals — Salaried customers show the highest conversion among occupation groups. Prioritise this segment in outbound sales campaigns.
- Improve Data Collection — Implement better data capture at the point of customer contact to reduce missing values in key fields like MonthlyIncome and DurationOfPitch for more accurate future analysis.